Lecture 19: The "S-Curve" (Control Rod Worth, Calibration, and Shadowing)

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

February 25, 2026

Introduction: The Misconception of the "Gas Pedal"

In a car, if you press the gas pedal 10% further, you generally expect a proportional increase in engine output. In a nuclear reactor, a Control Rod is **not linear**.

- Moving a rod 1 inch at the **bottom** of the core might add ≈ 0 reactivity.
- Moving a rod 1 inch in the **middle** of the core might add huge amounts of reactivity.
- Moving a rod 1 inch at the **top** of the core might add ≈ 0 reactivity again.

This non-linearity creates the famous **"S-Curve"** of rod worth. Understanding this curve is vital for predicting how the reactor will respond to operator actions and ensuring the core does not undergo an accidental power excursion.

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1 The Physics: Perturbation and Flux Weighting

Why does the rod effectiveness depend heavily on its position? We can explain this using **Perturbation Theory**.

The reactivity effect (ρ) of inserting a neutron-absorbing rod depends on two factors:

1. **The Local Flux (ϕ):** How many neutrons are physically present to be absorbed? If a region has no neutrons, inserting a rod there does nothing.
2. **The Adjoint Flux / Importance ($\phi^\dagger$):** How important is a neutron in that specific location to sustaining the overall chain reaction? In a simple, bare reactor, the "importance" of a location is directly proportional to the flux itself ($\phi^\dagger \approx \phi$).

Therefore, the "worth" of a control rod at a specific axial height z is proportional to the **square** of the unperturbed flux at that location:

$$\text{Worth}(z) \propto \phi(z)^2 \tag{1}$$

2 Differential and Integral Rod Worth

Let's consider a bare cylindrical reactor. The fundamental flux distribution along the vertical height (H) is a chopped sine wave:

$$\phi(z) = \phi_{max} \sin\left(\frac{\pi z}{H}\right)$$

2.1 Differential Rod Worth ($\frac{d\rho}{dz}$)

The Differential Rod Worth tells us how much reactivity we add or subtract *per inch* of rod movement. Based on our squared-flux rule:

$$\frac{d\rho}{dz} = C \sin^2\left(\frac{\pi z}{H}\right) \tag{2}$$

Where C is a scaling constant based on the macroscopic absorption cross-section of the rod material (e.g., Boron Carbide, Silver-Indium-Cadmium).

Physical Interpretation:

- **Core Edges ($z = 0, H$):** The flux approaches zero due to neutron leakage. Moving the rod near the very top or bottom has almost zero effect.
- **Core Center ($z = H/2$):** The flux is at its absolute maximum. The rod is "biting" into the highest population of neutrons. Moving the rod here causes rapid, massive changes in reactivity.

2.2 Integral Rod Worth (The S-Curve)

Operators need to know the **Total Reactivity** (ρ) inserted if they pull a rod from the fully inserted position ($z = 0$) to a specific withdrawal height (z). We find this by integrating the differential worth:

$$\rho(z) = \int_0^z \frac{d\rho}{dz'}\, dz' = \int_0^z C \sin^2\left(\frac{\pi z'}{H}\right) dz'$$

Using the trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$, we obtain the integral worth equation:

$$\boxed{\rho(z) = \rho_{total}\left[\frac{z}{H} - \frac{1}{2\pi}\sin\left(\frac{2\pi z}{H}\right)\right]} \tag{3}$$

This function generates the classic **S-shape** (Sigmoid) curve.

3 Core Distortion: Shadowing and Anti-Shadowing

The equations above assume a single rod in a "perfect" unperturbed flux. In reality, multiple rods interact with each other by physically reshaping the neutron flux profile.

- **Flux Depression:** A control rod acts like a black hole for thermal neutrons. The flux immediately surrounding the rod dips significantly.
- **Rod Shadowing:** If Rod A is inserted, it depresses the local flux. If Rod B is located right next to Rod A, Rod B is now sitting in a "neutron shadow." Because worth depends on ϕ^2, the worth of Rod B is *decreased* by the presence of Rod A.
- **Anti-Shadowing:** Conversely, if inserting Rod A pushes the overall core flux to the opposite side of the reactor, a Rod C located on that far side now experiences an artificially high flux. Rod C's worth is *increased* by the presence of Rod A.

To combat flux distortion and shadowing, reactors rarely move single rods. They move rods in symmetric, distributed groups called **Control Banks**.

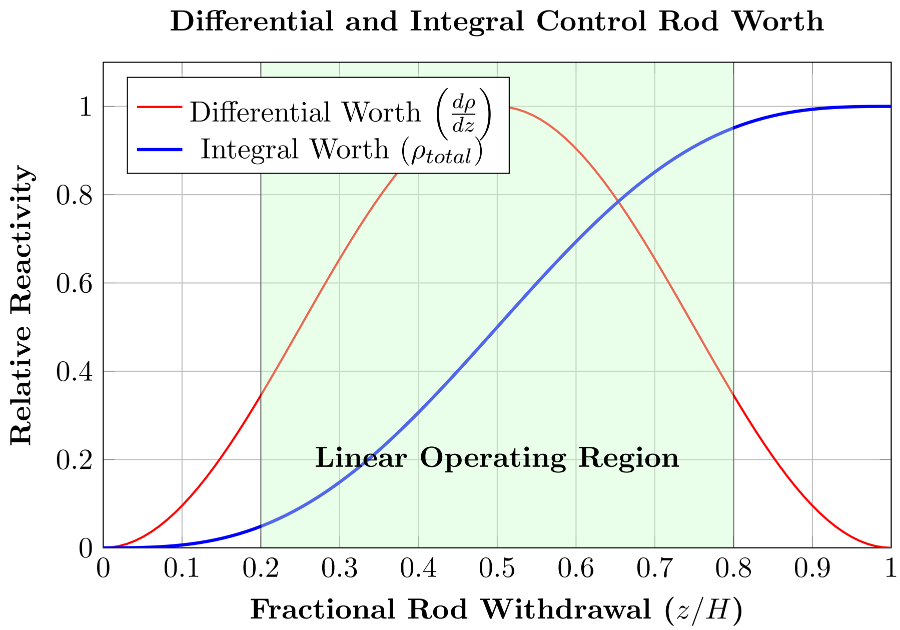

Figure 1: The mathematical relationship between rod position and reactivity. Operators prefer to keep rods in the shaded 20%-80% region where the S-Curve is roughly linear, ensuring predictable reactor response.

4 How Do We Measure It? (Rod Calibration)

You cannot just calculate rod worth; it must be experimentally measured for every new core loading. The most common technique is the **Positive Period Method**:

1. Establish critical steady-state at very low power.
2. Withdraw the test rod by a small increment (Δz). The reactor is now slightly supercritical.
3. Power begins to rise exponentially: $P(t) = P_0 e^{t/T}$.
4. Measure the Reactor Period (T), which is the time it takes for power to increase by a factor of e.
5. Use the **Inhour Equation** to convert T into actual Reactivity (ρ).
6. Add boric acid (or insert a different rod bank) to return the reactor to critical, and repeat the process for the next increment.

By summing the $\Delta\rho$ increments, we physically trace out the S-Curve.

5 Safety Implications and the SL-1 Accident

The ϕ^2 dependence has terrifying safety implications, famously demonstrated by the **SL-1 accident in 1961**.

The SL-1 was a small military prototype reactor in Idaho. It featured a massive central control rod (#9) that sat at the absolute peak of the radial flux distribution.

- The total delayed neutron fraction (β) is roughly 0.0065 (or $0.65\%\Delta k/k$). Injecting this amount of reactivity makes a reactor Prompt Critical.
- The central #9 rod in SL-1 was incredibly valuable, worth roughly $0.04\Delta k/k$ from top to bottom.
- During a maintenance procedure, an operator was supposed to lift the rod 4 inches (staying in the low-worth bottom tail of the S-curve).
- Due to a stuck mechanism, he manually yanked the rod, inadvertently withdrawing it approximately 20 inches. This pulled the rod straight through the maximum-worth center of the S-curve.
- He injected over $2\%\Delta k/k$ of reactivity (more than $3\times$ the prompt critical threshold) in a fraction of a second. The subsequent steam explosion destroyed the vessel and killed the three operators.

The Legacy - Stuck Rod Criterion: Modern power reactors are now legally required to be designed such that they will safely shut down (remain subcritical) even if the single most reactive control rod is completely ejected or stuck fully out of the core.

6 Control Rod Materials: Engineering Trade-offs

We know a rod needs a high macroscopic absorption cross-section (Σ_a), but nuclear engineers must also consider physical state, required cladding, lifespan, mechanical swelling, and cost. There is no "perfect" material.

Material	**Physical Form**	**Pros**	**Cons**
Boron Carbide (B_4C)	Ceramic pellets. **Must be cladded** in stainless steel.	Very high thermal σ_a. Cheap and lightweight.	(n,α) reaction produces Helium gas. Causes high internal pressure, requires a hollow gas plenum.
Silver-Indium-Cadmium (AIC)	Metal Alloy (80/15/5). **Must be cladded** in stainless steel.	Excellent epithermal resonance absorbers. Does not produce gas.	Heavy, expensive. Soft metal with poor corrosion resistance. Silver activates to Ag-110m (strong gamma).
Hafnium (Hf)	Solid Metal. **Unclad** (Directly exposed to coolant).	"Chain-absorbing" (isotopes remain absorbers after catching a neutron). Extreme corrosion resistance and strength.	Extremely heavy and highly expensive. Primarily reserved for Navy and high-performance reactors.

Table 1: Common Control Rod Materials in Thermal Reactors

6.1 The Helium Gas Problem (B_4C)

Boron is a fantastic absorber, but the B-10 reaction is:

$$^{10}\mathrm{B} + ^{1}\mathrm{n} \rightarrow ^{7}\mathrm{Li} + ^{4}\mathrm{He}$$

Over years of operation, a Boron control rod slowly fills with Helium gas. If the internal pressure exceeds the strength of the stainless steel cladding, the rod can swell, deform, and become physically stuck inside the core. To prevent this, B_4C rods are built with hollow "plena" (empty tube sections at the top) to give the gas room to expand.

6.2 The Hafnium Advantage

Unlike Boron or AIC, Hafnium is a refractory metal with chemical properties almost identical to Zirconium. It does not swell, and it does not corrode in high-temperature water. Consequently, Hafnium control blades do not need protective stainless steel cladding; they can be machined as solid metal components. Furthermore, when Hf-177 absorbs a neutron, it becomes Hf-178, which is *also* an excellent neutron absorber. This "chain-absorbing" property gives Hafnium an exceptionally long operational lifespan, making it the material of choice for nuclear submarines where refueling is exceptionally difficult.

7 Mechanical Design and Flux Shaping Strategies

How we physically shape and insert these absorbing materials depends on the thermal-hydraulic design of the reactor. Furthermore, engineers must account for what happens to the localized neutron flux when a rod is withdrawn.

- **PWR "Spider" Assemblies:** In a Pressurized Water Reactor (PWR), the water is a single-phase liquid. Control rods are bundled into clusters of roughly 24 thin rodlets attached to a central hub (a "spider"). These drop directly into hollow guide thimbles integrated into the fuel assembly lattice from above.
- **BWR Cruciform Blades:** In a Boiling Water Reactor (BWR), the fuel assemblies are enclosed in solid metal channel boxes to prevent boiling cross-flow. Therefore, BWRs use **cruciform (cross-shaped)** blades that slide into the gaps *between* four adjacent assemblies. They are driven hydraulically from the **bottom** to avoid the steam-drying equipment at the top of the vessel.
- **Flux Shaping and "Grey Rods":** Commercial reactors do not want severe, localized power spikes when rods are moved. To manage the axial power distribution smoothly:
  - **Axial Grading:** The absorber material is often varied along the length of the rod. The tip may contain a weaker absorber (or a lower enrichment of B-10) than the main body to "soften" the neutron shadow and prevent a sharp power peak in the fuel immediately below the rod.
  - **Grey Rods:** Advanced PWRs utilize specialized "Grey Rods." Unlike standard "Black Rods" designed to heavily absorb neutrons and shut the reactor down, grey rods use weaker absorbing materials (like Inconel or stainless steel). They are used purely to gently shape the axial flux profile and control Xenon transients without causing massive local flux depressions.

- **Follower Rods (Research & Test Reactors):** In high-flux test reactors, pulling a solid control rod out of the core leaves an empty channel that immediately fills with water. This sudden block of pure moderator causes a dangerous localized thermal flux spike. To prevent this, rods are built with "followers"—a solid section of low-absorbing material (like aluminum or beryllium) attached to the bottom of the absorber. As the absorber is pulled up, the follower slides into the core to displace the water and keep the moderation profile perfectly flat.

8 Case Study in Design Failure: The RBMK Control Rod

The ultimate safety mechanism of any nuclear reactor is the SCRAM system (the rapid insertion of control rods). A fundamental axiom of nuclear engineering is that the activation of the emergency shutdown system must introduce **strictly negative reactivity** at all times and at all points in the core.

The Soviet RBMK-1000 reactor (the design used at Chernobyl) famously violated this axiom. The design of its control rods serves as a textbook object lesson in how optimizing a system for steady-state economic efficiency can completely compromise its transient safety.

8.1 The Economic Motivation: The "Follower" Concept

In the RBMK, the primary neutron moderator is a massive structure of solid graphite blocks, while ordinary light water (H_2O) flows through pressure tubes to cool the fuel.

From a neutronics perspective, light water is a mild neutron absorber ($\sigma_a \approx 0.66$ barns for H_2O), whereas graphite is highly transparent to neutrons ($\sigma_a \approx 0.003$ barns).

When a standard control rod (made of Boron carbide) is pulled entirely out of the core, the empty channel fills with coolant water. This column of water acts as a parasitic poison, absorbing neutrons and lowering the fuel efficiency of the reactor. To solve this, Soviet engineers attached a graphite "displacer" (or follower) to the bottom of the boron control rod.

The logic was simple: when the boron is pulled out, the graphite displacer is pulled into the core, physically displacing the water. Because graphite absorbs far fewer neutrons than water, this saved neutrons and improved the economic efficiency of the reactor.

8.2 The Geometric Flaw

The fatal error lay in the exact physical dimensions of the rod assembly relative to the reactor core.

- **Core Height:** 7.0 meters.
- **Boron Absorber Height:** 7.0 meters.
- **Graphite Displacer Height:** 4.5 meters.
- **Telescoping Linkage:** 1.25 meters (connecting the boron to the graphite).

Because the graphite displacer was shorter than the core, its resting position when "fully withdrawn" was incredibly dangerous. When the operator pulled the rod to its upper limit, the 4.5-meter graphite displacer sat exactly in the middle of the 7-meter core.

Crucially, this left a **1.25-meter column of water at the very bottom of the core channel** (and another 1.25 meters of water at the top).

8.3 The Transient: The "Positive SCRAM"

When the operators pressed the AZ-5 button to initiate an emergency SCRAM, the rod assembly began to move downward.

1. At the *top* of the core, the boron absorber entered, replacing water. This added negative reactivity ($\Delta\rho < 0$), exactly as a control rod should.
2. At the *bottom* of the core, the graphite displacer moved downward into the 1.25-meter column of water.

For the first few seconds of the SCRAM, the bottom of the control channel was physically pushing out a neutron absorber (water) and replacing it with a fantastic neutron moderator (graphite). This action locally increased the thermal utilization factor (f), causing a massive, localized spike of **positive reactivity** ($\Delta\rho > 0$) at the bottom of the reactor.

8.4 The Kinetic Flaw: Slow Insertion Speed

A localized positive reactivity spike is dangerous, but it might have been survivable if the boron absorber followed instantly to shut down the chain reaction. This brings us to the mechanical drive system.

In Western commercial reactors (like PWRs and BWRs), control rods drop via gravity or hydraulic pressure into the core in roughly **2 to 4 seconds**.

The RBMK rods were driven by electric motors geared to move at an agonizingly slow speed of 0.4 meters per second. A full SCRAM from the top of the core took **18 to 20 seconds**.

Because the rods moved so slowly, the graphite tips sat in the active lower core, displacing water and adding positive reactivity for several seconds. This was more than enough time for the lower fuel assemblies to experience a prompt critical power excursion.

8.5 The Xenon Trap (The Catalyst)

As you learned in the last lecture, Xenon poisoning following a power reduction can force operators to withdraw control rods to maintain criticality.

On the night of the Chernobyl disaster, a massive Xenon transient (the "Iodine Pit") had essentially poisoned the core. To keep the reactor running, the operators withdrew almost every single control rod to its absolute upper physical limit.

This meant that virtually all of the 4.5-meter graphite displacers were perfectly aligned in the center of the core, with 1.25-meter columns of water sitting directly beneath them. When the AZ-5 button was finally pressed, nearly 200 graphite tips entered the lower water columns simultaneously, inserting a massive spike of positive reactivity into the lower core. This initial power surge caused the remaining cooling water to flash to steam, which (due to the reactor's positive void coefficient) exacerbated the reactivity insertion in a vicious feedback loop. The resulting prompt-critical excursion rapidly melted the fuel channels, physically jamming the rods before the boron could ever reach the high-flux region.

Engineering Takeaways:

1. **System Constraints:** Never optimize for steady-state economics at the expense of transient safety.

2. **Monotonic Safety Action:** A safety shutdown system must be strictly monotonic—it must *only* add negative reactivity, regardless of rod position or core state.
3. **Response Time:** The mechanical speed of the safety system must outpace the kinetic speed of the neutronic excursion it is designed to prevent.

9 Summary

- **Non-Linearity:** Control rods are not linear dials; their worth scales with the local flux squared (ϕ^2).
- **Differential Worth:** Follows a chopped $\sin^2(z)$ shape, peaking exactly in the center of the core.
- **Integral Worth:** Follows an S-Curve.
- **Shadowing:** Rods influence the worth of other rods by squishing and pushing the spatial neutron flux.
- **Calibration:** Conducted carefully using the Reactor Period measurement.

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